

Article

Early-Career Choices in the Era of Generative AI: Human Capital Depreciation, Career Anxiety, and Occupational Adaptation Among Fresh Graduates

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Abstract

As generative artificial intelligence (GenAI) becomes most capable at precisely the entry-level tasks from which early careers have traditionally been built, fresh graduates must make their first occupational choices under conditions no previous cohort has faced. Drawing on socio-technical systems theory and a cybernetic control perspective, this study investigates how graduates' perceived human capital depreciation translates into AI-related career anxiety and, in turn, into occupational adaptation or career avoidance, and which combinations of conditions govern this process. Using a two-wave survey of 1,247 Chinese fresh graduates within three years of graduation (914 matched across waves, with behavioral outcomes measured six months after their antecedents), we employ confirmatory factor analysis, bootstrapped conditional-process models, and fuzzy-set qualitative comparative analysis (fsQCA). The study finds that (1) perceived human capital depreciation significantly raises AI-related career anxiety ($\beta = 0.487$), and perceived employability support buffers this first stage, reducing the pass-through from 0.64 to 0.33; (2) career anxiety exhibits an inverted U-shaped relationship with occupational adaptation, with a turning point 0.37 standard deviations above the mean anxiety level, beyond which 38.5 percent of graduates already fall, whereas career avoidance rises monotonically; (3) GenAI literacy displaces the turning point outward from 0.07 to 0.66 standard deviations, extending the range over which anxiety remains productive; and (4) the fsQCA identifies a single sufficient configuration for high adaptation, the conjunction of anxiety, literacy, and support (consistency 0.892), with no necessary condition and no consistent configuration for its negation, so anxiety alone never suffices. This study deepens the systems understanding of GenAI-era workforce development by modeling career anxiety as the error signal of an adaptive feedback loop, and offers practical guidance for universities,

employers, and policymakers seeking to convert a record cohort's anxiety into adaptation rather than avoidance.

Keywords: generative artificial intelligence; fresh graduates; career anxiety; human capital depreciation; occupational adaptation; socio-technical systems; complex adaptive systems; moderated mediation; fuzzy-set qualitative comparative analysis; workforce development

1. Introduction

Every technological transition redistributes opportunity across the age distribution, but the present one is unusual in where it strikes first. Generative artificial intelligence performs best on the codifiable, language-intensive, well-specified tasks that have traditionally been assigned to the newest members of the professions [1–3]. Field and quasi-experimental evidence now shows entry positions contracting where the technology is adopted: junior-heavy hiring falls in exposed occupations [4,5], the productivity gains of the technology accrue disproportionately to novices while the demand for novices weakens [6,7], and employers increasingly describe entry roles in terms of supervising machine output rather than producing it [8,9]. For the roughly twelve million students who now graduate from Chinese higher education each year [10,11], these are not abstractions. They are the conditions under which the first occupation, the first investments in skill, and the first revisions of both must be chosen.

Economics has begun to measure what the technology does to the demand side of this market [12–14]. Much less is known about the supply side: how the entrants themselves perceive the depreciation of their just-acquired human capital, what that perception does to them emotionally, and when the emotion converts into behavior that improves their position rather than behavior that worsens it. The question matters because early-career scarring is among the most persistent phenomena in labor economics: cohorts that stumble at entry carry lower earnings and worse matches for a decade or more [15–18]. If the first cohort of the generative-AI era responds to depreciation signals with paralysis rather than adaptation, the technology's distributional effects will be amplified by the very psychology of those it displaces.

This study asks three questions. First, how does the perception that generative artificial intelligence is depreciating one's human capital translate into AI-related career anxiety? Second, when does that anxiety mobilize occupational adaptation, and when does it tip into avoidance? Third, which combinations of psychological and contextual conditions are jointly sufficient for adaptation? The questions are posed from an explicitly systems-theoretic position, which is what distinguishes this paper from the emerging literature on artificial-intelligence anxiety [19–21]. The graduate is modeled as an adaptive agent inside a socio-technical labor system [22,23]: the depreciation percept is the input, anxiety is the error signal of a control loop that compares the career one has with the career one expected [24,25], and reskilling, occupational repositioning or withdrawal are the actions through which the loop closes. Three properties follow from this framing, and each is testable. A control signal should transmit the input (mediation); an error signal that saturates should stop producing corrective action at high amplitude

(nonlinearity); and the action should depend on the actuators available to the agent, namely capability and support (conjunction). The empirical design maps one to one onto these properties.

Evidence comes from a two-wave survey of Chinese graduates within thirty-six months of graduation, with perceptions and psychological states measured at the first wave and adaptation behaviors measured six months later, so that the outcomes are separated from their hypothesized antecedents in time. The analysis combines a confirmatory measurement model, bootstrapped conditional-process estimation with an explicitly curvilinear second stage [26–28], and fuzzy-set qualitative comparative analysis [29–31], the latter because a systems argument about conjunction cannot be tested by adding interaction terms alone [32].

Four results organize the paper. First, perceived human capital depreciation converts into career anxiety at $\beta = 0.487$ and the conversion is buffered by perceived employability support: the pass-through is 0.64 one standard deviation below the mean of support and 0.33 one standard deviation above it. Second, the anxiety-adaptation relationship is an inverted U, not a slope. Adaptation peaks at 0.37 standard deviations of anxiety, and 38.5 percent of the cohort already sits beyond the peak, in the region where additional anxiety predicts less adaptation; avoidance, by contrast, rises monotonically across the entire observed range. Third, generative-AI literacy does not merely raise adaptation, it moves the turning point: from 0.07 standard deviations among low-literacy graduates to 0.66 among high-literacy ones, so that literacy extends the range over which anxiety remains productive. Fourth, the configurational analysis returns exactly one sufficient recipe for high adaptation, the conjunction of anxiety with literacy and support (consistency 0.892, coverage 0.406); no single condition is necessary, no configuration lacking either literacy or support passes the sufficiency thresholds, and no consistent recipe exists for the negation. Equifinality, the default expectation of configurational research, fails here in a theoretically informative way: there is one road to adaptation, and anxiety alone is never it.

The contribution is threefold. To the economics and management literature on generative artificial intelligence and work [6,12,33], the paper adds the supply-side psychology of the displaced margin: the first individual-level evidence that the entry cohort's behavioral response to depreciation is nonmonotonic, which changes what aggregate adjustment models should assume about retraining uptake. To the artificial-intelligence anxiety literature [19,20,34], it contributes the systems treatment: anxiety modeled and estimated as the error signal of a feedback loop, with the saturation, the actuator-dependence and the conjunction that the control-theoretic reading predicts [24,25,35]. To the special issue's question of how generative artificial intelligence reshapes workforce development as a system [36,37], it offers a concrete demonstration that the productive unit of analysis is neither the individual nor the technology but the loop that couples perception, emotion, capability and institutional support, and that policy which addresses only one element of the loop, such as reassuring messaging that damps the signal, can be worse than policy which completes it.

The remainder of the paper is organized as follows. Section 2 develops the theory and seven hypotheses. Section 3 describes the sample, the measures and the analysis strategy. Section 4 reports the measurement model, the structural estimates, the conditional-process analysis, the heterogeneity and

robustness batteries and the configurational results. Section 5 discusses theoretical and practical implications, Section 6 acknowledges limitations, and Section 7 concludes.

2. Theory and Hypotheses

2.1. Fresh Graduates as Adaptive Agents in a Socio-Technical Labor System

A labor market entrant is not a passive recipient of technological change; they are an agent embedded in a system of universities, employers, technologies and institutions whose joint behavior determines early-career outcomes [22,23,38]. Systems thinking supplies three ideas that discipline the analysis [39]. The first is the feedback loop [40,41]: perceptions drive emotions, emotions drive actions, and actions feed back into the perceptions and the objective position from which the next round of choices is made. The second is the error signal of cybernetic control theory [24,42]: an adaptive system monitors the discrepancy between its present trajectory and its reference trajectory, and the affective correlate of that discrepancy is what mobilizes correction. The third is requisite capability, the behavioral cousin of Ashby's law [25]: a control signal produces regulation only if the regulator commands actions varied enough to match the disturbance. An anxious graduate who cannot use the technology and has no institutional support has a loud error signal wired to no actuator.

The disturbance in question is well documented on the demand side. Generative models write, summarize, draft, code and analyze at a level that overlaps heavily with entry-level professional work, the modern echo of the routine-task displacement long documented for earlier automation [1,3,43,44], and the overlap is largest precisely in the occupations that hire educated entrants [2,45]. Adoption evidence links the technology to falling junior hiring in exposed firms and platforms [4,5,46], and experimental evidence shows the technology compressing the novice-expert productivity gradient that entry-level jobs exist to climb [6,7,47,48] — even while economy-wide employment effects remain modest so far [49], which is precisely why the entry margin, where the effects concentrate [50], deserves separate study. Chinese firm-level evidence points the same way [51–55]. What has not been studied is the perception of this disturbance by those on its receiving end, and the system through which perception becomes choice.

2.2. Perceived Human Capital Depreciation and Career Anxiety

Human capital depreciates when the market value of a skill stock falls, whether through wear, through vintage effects or through technological displacement of the tasks the skills perform [56–58]. The loss lands hardest on entrants because early careers are built by doing: skills are accumulated on the job [59], weighted and priced by the tasks a position offers [60], and advanced through internal ladders whose first rung is exactly the work the technology absorbs [61]. What matters for behavior is not the depreciation itself but its perception. A graduate observes the capability of generative systems in their own domain, compares it with what they were trained to do, and forms a judgment about how fast the training is losing value [21,62]. In appraisal-theoretic terms this is a primary appraisal of threat to a central resource, the career [63]; in conservation-of-resources terms it is anticipated resource loss, the most potent known elicitor of stress [64,65]. Both traditions, and the cybernetic reading in which the percept is a widening discrepancy

between expected and likely trajectory [24], predict the same first link: perceived depreciation should convert into career-specific anxiety, the unpleasant activation state focused on one's occupational future [19,20,66]. Accordingly:

H1. *Perceived human capital depreciation is positively associated with AI-related career anxiety.*

2.3. The Dual Response: Adaptation and Avoidance

Anxiety is an activation state, and activation is behaviorally ambidextrous. The coping literature distinguishes problem-focused responses, which attack the stressor, from emotion-focused and avoidant responses, which attack the feeling [63]. In the career domain the problem-focused response is occupational adaptation: enrolling in structured reskilling, pursuing credentials, deliberately practicing the new technology, and repositioning toward tasks the technology complements rather than substitutes [67,68]. The avoidant response is withdrawal: postponing career decisions, avoiding information about the technology, and contemplating exit from the chosen field [20,21].

The relationship between anxiety and the problem-focused response should not be monotone. At low levels, anxiety supplies the activation that overcomes the inertia of costly investment: this is the mobilizing arm familiar from the oldest result in behavioral activation research [69] and from conservation-of-resources theory, in which moderate loss threat triggers resource investment [65]. At high levels, three mechanisms bend the curve down. Cognitively, high anxiety narrows attention and consumes the working memory that planning a retraining program requires. Motivationally, extreme threat triggers defensive avoidance rather than approach [63]. And in resource terms, graduates deep in loss enter the defensive spiral in which conserving what remains dominates investing in what could be [65]. The general form of this argument, that a productive input becomes counterproductive past an interior optimum, is the too-much-of-a-good-thing effect [70,71], and in control-theoretic language it is saturation: an error signal beyond the actuator's range ceases to improve regulation [24,42]. The avoidant response carries no such interior optimum; whatever anxiety remains unconverted into action discharges as escape, monotonically. Hence:

H2. *AI-related career anxiety has an inverted U-shaped relationship with occupational adaptation: adaptation rises with anxiety up to a turning point and declines beyond it.*

H3. *AI-related career anxiety is positively and monotonically associated with career avoidance.*

2.4. Anxiety as the Conduit

If anxiety is the error signal through which the depreciation percept reaches behavior, then depreciation should influence both responses through anxiety, and the sign and size of the transmitted effect should inherit the curvature of the second stage. Formally, the indirect effect of depreciation on adaptation is the product of the first-stage slope and the local second-stage slope, and because the latter changes sign at the turning point, the same depreciation signal that mobilizes a calm graduate demobilizes an already anxious one [26,27]. This is a stronger and more falsifiable claim than simple mediation, and it is the signature prediction of the control-loop reading:

H4a. *AI-related career anxiety transmits the effect of perceived human capital depreciation to occupational adaptation, and the transmitted effect declines with the level of anxiety, turning negative beyond the turning point.*

H4b. *AI-related career anxiety transmits a positive effect of perceived human capital depreciation to career avoidance.*

2.5. Boundary Conditions: Literacy and Support

What separates the graduates whose anxiety converts into adaptation from those whose anxiety converts into escape? The systems answer is the actuator set. The first actuator is generative-AI literacy: the self-efficacious command of the technology itself, the ability to prompt, verify, and integrate machine output into one's own work [19,62,72]. Literacy raises the expected return on every unit of adaptive effort, because a graduate who can already use the tool converts a reskilling course into complementarity faster than one who cannot [73,74]. In the geometry of the inverted U, a higher payoff to effort holds the upward arm positive for longer, displacing the turning point outward [28]; the same command of the tool should drain the avoidance channel, because escape loses its appeal when engagement is feasible. The second actuator is perceived employability support: the graduate's assessment that university career services, employer training provision and their professional network stand behind their transition [75,76]. Support operates earlier in the loop. It reframes the depreciation percept as a solvable problem, so the same objective signal generates less anxiety in the first place, the classic buffering position of the stress literature [63,65]. Accordingly:

H5a. *Generative-AI literacy moderates the second stage: it raises the marginal payoff of anxiety for adaptation and displaces the turning point outward.*

H5b. *Generative-AI literacy weakens the positive association between anxiety and career avoidance.*

H6. *Perceived employability support weakens the first stage: the association between perceived depreciation and anxiety is smaller at higher support.*

2.6. A Configurational View: Necessity and Sufficiency

Interaction terms test whether conditions modify each other at the margin; they do not test the stronger systems claim that outcomes are produced by configurations, combinations of conditions that succeed or fail as wholes [30,35,77]. Complex adaptive systems typically exhibit equifinality, multiple distinct paths to the same outcome [29,78], and the default expectation of configurational research in the information-systems and management traditions is that several recipes will surface [31,32]. The control-loop argument, however, points the other way. If adaptation requires a signal (anxiety at a workable level), an actuator (literacy) and an enabling environment (support) simultaneously, then the recipes should collapse toward a single conjunction, and no individual ingredient should be sufficient or strictly necessary on its own. The configurational analysis therefore tests two claims against each other:

H7a. *No single psychological or contextual condition is necessary for high occupational adaptation.*

H7b. *Multiple distinct configurations of depreciation, anxiety, literacy, support and exposure are sufficient for high occupational adaptation (equifinality).*

The hypotheses jointly describe one mechanism: a percept converted into an error signal (H1), the signal converted into corrective action within a bounded range (H2, H4a) or discharged as escape (H3, H4b), the range extended by capability (H5a, H5b), the signal damped at source by support (H6), and the whole loop operating conjuncturally rather than additively (H7a, H7b). No path diagram is needed; the mechanism is the sequence of estimating equations in Section 3.3.

3. Materials and Methods

3.1. Sample and Procedure

The population of interest is mainland Chinese university graduates within thirty-six months of graduation, the entrants whose early careers coincide with the diffusion of generative artificial intelligence. Respondents were recruited through a professional online panel supplemented by university alumni associations across provinces, with quotas on gender, degree level and city tier to approximate the national graduate profile [10,79]. The first wave, fielded in September 2025, measured perceptions, psychological states, moderators and controls. The second wave, fielded six months later in March 2026, measured the two behavioral outcomes. Attention checks, minimum-duration screens and straight-lining filters were applied, yielding 1,247 valid first-wave responses, of which 914 were matched at the second wave, a retention rate of 73.3 percent. Table 1 profiles the sample and compares retained with lost respondents: the smallest attrition p-value across observables is 0.095, and anxiety itself does not predict attrition ($p = 0.479$), though inverse-probability weights are used in robustness regardless. The median monthly income of employed respondents is 6,300 yuan, in line with published graduate salary distributions [10]. Respondents span 20 occupation groups.

Table 1. Sample profile and attrition analysis.

Characteristic	Time 1 (N = 1247)	Retained (N = 914)	Lost (N = 333)	p
Female (%)	52.6	52.8	52.0	0.780
Age (years)	23.6	23.6	23.6	0.461
Master's degree (%)	18.2	18.4	17.7	0.787
STEM major (%)	40.4	40.3	40.8	0.854
Tier-1 city (%)	31.9	32.6	30.0	0.384
Months since graduation	18.7	18.7	18.6	0.850
Employed (%)	72.7	74.0	69.1	0.095
Parent with tertiary education (%)	28.4	28.0	29.4	0.626
Internship experience (%)	65.0	65.3	64.3	0.731
Depreciation items, mean	4.0	4.0	4.0	0.534
Anxiety items, mean	4.0	3.9	4.0	0.479

Note: Continuous characteristics are means; binary characteristics are percentages. The final column reports the p-value of a Welch two-sample test of the difference between retained and lost respondents. Depreciation and anxiety rows are unweighted means of the raw seven-point items at the first wave. Source: authors' own calculations.

3.2. Measures

All perceptual constructs use seven-point Likert items (1 = strongly disagree, 7 = strongly agree), adapted from validated instruments to the generative-AI context and translated by the standard back-translation procedure. Perceived human capital depreciation (PHCD, four items) captures the judgment that one's degree skills are losing labor market value as generative systems improve, adapting skill-obsolescence measures [57,58] to the technology (sample item: "The skills I acquired in my degree are losing value as generative AI systems improve"). AI-related career anxiety (ANX, four items) adapts the job-replacement dimension of the artificial-intelligence anxiety scale [19] to the career frame of reference ("Thinking about what AI means for my career path makes me tense"). Generative-AI literacy (LIT, four items) measures self-efficacious command of the technology [62,72]: prompting effectively, verifying output, integrating it into one's own work, and keeping pace with new releases. Perceived employability support (SUP, four items) measures the graduate's assessment of university career services, employer training provision and network support [75]. Occupational adaptation (ADP, four items, second wave) records behavior over the intervening six months: enrollment in structured reskilling, credential pursuit, deliberate practice with generative tools, and movement toward tasks that complement the technology [67,68]. Career avoidance (AVD, three items, second wave) records postponed career decisions, avoidance of AI-related information, and contemplated exit from the field [20]. A three-item marker construct, theoretically unrelated aesthetic preference, is carried for method-bias diagnostics [80].

Occupational exposure (EXPO) is the one non-perceptual condition: each respondent's current or target occupation group is scored for generative-AI exposure by aggregating task-level language-model applicability to the occupation level, following the construction logic of the exposure literature [1,2], and standardized over the sample (sample mean of the raw score 0.58, standard deviation 0.18). As a behavioral cross-check on the adaptation index, the second wave also records hours of structured reskilling in the past six months. Controls comprise gender, age, degree level, STEM major, city tier, months since graduation, employment status, parental tertiary education and internship experience, the standard determinants of early-career outcomes [15,16].

3.3. Analysis Strategy

The analysis proceeds in four blocks, each matched to a property of the hypothesized control loop. The first block is the measurement model: a maximum-likelihood confirmatory factor analysis of the six substantive constructs, with reliability, convergent and discriminant validity assessed by the conventional criteria [81–83], measurement invariance tested across gender and major, and common-method risk assessed by design (temporal separation of the outcomes), by the Harman diagnostic, by an equal-loading unmeasured method factor and by the marker technique [80,84,85]. Bartlett factor scores carry the constructs into the structural block; factor-score regression of this form is consistent for the structural parameters under a correctly specified measurement model [86], and unit-weighted composites replicate every result in robustness.

The second block estimates the three structural equations

$$ANX_i = a_0 + a_1 PHCD_i + a_2 SUP_i + a_3 (PHCD_i \times SUP_i) + \gamma'_1 X_i + \varepsilon_{1i} \quad (1)$$

$$ADP_i = b_0 + b_1 ANX_i + b_2 ANX_i^2 + b_3 (ANX_i \times LIT_i) + b_4 LIT_i + b_5 SUP_i + b_6 PHCD_i + \gamma'_2 X_i + \varepsilon_{2i} \quad (2)$$

$$AVD_i = c_0 + c_1 ANX_i + c_2 (ANX_i \times LIT_i) + c_3 LIT_i + c_4 SUP_i + c_5 PHCD_i + \gamma'_3 X_i + \varepsilon_{3i} \quad (3)$$

with all continuous variables standardized so that interaction terms are interpretable in the conventional way [87], heteroskedasticity-consistent (HC3) standard errors, and occupation-clustered inference in robustness. H2 is not accepted from the sign of b_2 alone: following the now-standard critique of quadratic curve-fitting [28,88], the composite test requires a significantly positive slope at the low extreme of the anxiety distribution and a significantly negative slope at the high extreme, and the turning point

$$\tau(l) = \frac{-b_1 + b_3 l}{2b_2} \quad (4)$$

is reported with a Fieller confidence interval at each level l of literacy, so that H5a is a statement about the displacement of τ . The third block is the conditional-process analysis [26,27,89,90]. Because the second stage is curvilinear, the indirect effect of depreciation on adaptation is the product of the conditional first-stage slope and the local second-stage derivative,

$$\omega(m, w, l) = (a_1 + a_3 w)(b_1 + 2b_2 m + b_3 l) \quad (5)$$

evaluated at chosen levels of anxiety m , support w and literacy l , with 95 percent percentile confidence intervals from 5,000 bootstrap resamples; the indices of moderated mediation are $a_3 b_1$ for support and $a_1 b_3$ for literacy [27], and $2a_1 b_2$ indexes the curvature of the transmitted effect itself. The avoidance channel is $a_1(c_1 + c_2 l)$.

The fourth block is the fuzzy-set qualitative comparative analysis [29,30,78]. The five conditions (PHCD, ANX, LIT, SUP, EXPO) and the outcome (ADP) are calibrated by the direct method, with full membership, crossover and full non-membership anchored at the 95th, 50th and 5th sample percentiles [31],

$$\mu(x) = \frac{1}{1 + e^{-z(x)}} \quad (6)$$

where the log-odds score is $z(x) = 3(x - c)/(f - c)$ at or above the crossover and $z(x) = 3(x - c)/(c - o)$ below it, with f , c and o the full-membership, crossover and non-membership anchors. Necessity and sufficiency are judged by consistency and coverage,

$$Cons(X \leq Y) = \frac{\sum_i \min(x_i, y_i)}{\sum_i x_i} \quad (7)$$

$$Cov(X, Y) = \frac{\sum_i \min(x_i, y_i)}{\sum_i y_i} \quad (8)$$

with a truth-table frequency threshold of ten cases, raw consistency of at least 0.80 and proportional reduction in inconsistency of at least 0.70 [31,32]. The intermediate solution admits only remainders consistent with the directional expectations that depreciation, literacy, support and exposure, where present, favor adaptation, with no expectation imposed on anxiety; core conditions are those that survive into the parsimonious solution [30]. The entire pipeline, from raw responses to every number in this paper, is released as documented code, and all bootstrap and resampling seeds are fixed in the scripts.

4. Results

4.1. Measurement Model

Table 2 reports the confirmatory factor analysis. The six-construct model fits well: $\chi^2(215) = 277.35$ ($\chi^2/df = 1.29$), CFI = 0.994, TLI = 0.993, RMSEA = 0.018, SRMR = 0.022, comfortably inside the conventional cutoffs [82]. The smallest standardized loading is 0.731, the smallest Cronbach alpha and composite reliability are 0.79, and the smallest average variance extracted is 0.556, above the 0.50 benchmark, establishing convergent validity [81].

Table 2. Constructs, items, and convergent validity.

Construct and items	Loading	α	CR	AVE
Perceived human capital depreciation (PHCD)				
phcd1	0.800	0.873	0.873	0.633
phcd2	0.780			
phcd3	0.810			
phcd4	0.791			
AI-related career anxiety (ANX)				
anx1	0.822	0.889	0.889	0.667
anx2	0.823			
anx3	0.806			
anx4	0.817			
Generative-AI literacy (LIT)				
lit1	0.795	0.865	0.865	0.616
lit2	0.790			
lit3	0.771			
lit4	0.784			
Perceived employability support (SUP)				
sup1	0.787	0.855	0.856	0.597
sup2	0.767			
sup3	0.767			
sup4	0.770			
Occupational adaptation (ADP)				
adp1	0.768	0.845	0.845	0.577
adp2	0.762			
adp3	0.757			
adp4	0.750			
Career avoidance (AVD)				
avd1	0.758	0.790	0.790	0.556
avd2	0.749			
avd3	0.731			

Note: Loadings are standardized; α is Cronbach’s alpha. Maximum-likelihood confirmatory factor analysis on the matched sample (N = 914). Model fit: $\chi^2(215) = 277.35$, CFI = 0.994, TLI = 0.993, RMSEA = 0.018, SRMR = 0.022. CR is composite reliability; AVE is average variance extracted. All loadings are significant at the 1% level. Source: authors’ own calculations.

Table 3 assesses discriminant validity from both directions. The square root of each construct’s average variance extracted (the smallest is 0.746) exceeds its largest latent correlation (the largest anywhere is 0.565), satisfying the Fornell-Larcker criterion, and the largest heterotrait-monotrait ratio is 0.566, well below the conservative 0.85 bound [83].

Table 3. Discriminant validity.

Construct	PHCD	ANX	LIT	SUP	ADP	AVD
PHCD	0.796	0.545	0.065	0.026	0.078	0.344
ANX	0.545	0.817	0.002	0.189	0.166	0.566
LIT	−0.064	−0.002	0.785	0.005	0.327	0.168
SUP	0.026	−0.190	0.004	0.773	0.211	0.219
ADP	0.079	0.166	0.327	0.212	0.760	0.022
AVD	0.343	0.565	−0.167	−0.219	−0.023	0.746

Note: Diagonal: square root of the average variance extracted. Below the diagonal: latent correlations from the confirmatory model. Above the diagonal: heterotrait–monotrait ratios. The Fornell–Larcker criterion and the HTMT < 0.85 criterion are both satisfied for every pair. Source: authors’ own calculations.

Table 4 completes the measurement block. Metric invariance holds across gender and across STEM and non-STEM majors: constraining loadings to equality changes the comparative fit index by less than 0.001 in both cases (chi-square difference $p = 0.543$ and 0.545), so the structural comparisons of Section 4.8 rest on equivalent measurement. Method bias is bounded from three directions: the outcomes were measured six months after their antecedents by design [84]; the first factor of the item pool carries 22.4 percent of the variance, far below the forty-percent alarm level; an equal-loading unmeasured method factor absorbs 13.1 percent of item variance, within the range regarded as unproblematic [85]; and the largest correlation between the marker construct and any substantive score is 0.039 [80]. Method variance exists, as it always does in self-report, but it is small and, as Section 4.9 shows, partialling it out moves nothing.

Table 4. Measurement invariance and common-method diagnostics.

Model	χ^2	df	CFI	$\Delta\chi^2$ (Δdf)	p
Gender: configural	519.8	430	0.991		
Gender: metric	541.4	453	0.991	21.6 (23)	0.543
Major: configural	508.4	430	0.992		
Major: metric	530.0	453	0.992	21.6 (23)	0.545
Harman single factor: first-factor share (%)	22.4				
Method factor: variance share (%)	13.1				

Note: Configural and metric two-group models estimated by maximum likelihood; the metric model constrains loadings to equality across groups. The Harman row reports the share of item variance carried by the first principal component; the method-factor row reports the variance share of an equal-loading unmeasured method factor added to the measurement model. Source: authors’ own calculations.

4.2. Descriptive Statistics

Table 5 reports means, standard deviations and correlations of the factor scores and exposure. The raw correlations already carry the paper’s outline: depreciation with anxiety at 0.480, exposure with depreciation at 0.492, and anxiety with adaptation at only 0.144 — the small linear correlation that an inverted U leaves behind when the two arms cancel. No pairwise correlation approaches the level at which collinearity would trouble the structural estimates.

Table 5. Descriptive statistics and correlations of the factor scores.

Variable	Mean	SD	PHCD	ANX	LIT	SUP	ADP	AVD	EXPO
PHCD	0.000	1.000	1.000						
ANX	0.000	1.000	0.480	1.000					
LIT	0.000	1.000	−0.056	−0.002	1.000				
SUP	0.000	1.000	0.022	−0.166	0.003	1.000			
ADP	0.000	1.000	0.068	0.144	0.280	0.180	1.000		
AVD	0.000	1.000	0.285	0.474	−0.139	−0.180	−0.019	1.000	
EXPO	0.000	1.000	0.492	0.263	0.029	0.012	0.034	0.153	1.000

Note: Bartlett factor scores, standardized; EXPO is the standardized occupational exposure score. Lower triangle: Pearson correlations (N = 914). Correlations of magnitude 0.07 or larger are significant at the 5% level. Source: authors’ own calculations.

4.3. First Stage: Depreciation into Anxiety (H1, H6)

Table 6 estimates Equation (1) in progressive columns. Alone, perceived depreciation carries anxiety with a standardized coefficient near one half; controls barely move it (column (2)), and support enters negatively in column (3). Column (4) adds the interaction: the pass-through is 0.487 (SE 0.027) at mean support, support lowers anxiety directly (−0.171), and the interaction is −0.154 (p < 0.001). Evaluated one standard deviation below and above mean support, the same depreciation percept converts at 0.64 versus 0.33 — support cuts the first stage nearly in half. Column (5) repeats the specification on the full first-wave sample (N = 1,247) with practically identical coefficients (pass-through 0.455), so panel attrition does not shape the first stage. H1 and H6 are supported. In control-loop terms, support does not act on the signal’s consequences; it recalibrates the sensor.

Table 6. First stage: perceived depreciation into career anxiety (H1, H6).

	(1)	(2)	(3)	(4)	(5)
PHCD	0.480*** (0.028)	0.486*** (0.029)	0.490*** (0.028)	0.487*** (0.027)	0.455*** (0.024)
SUP			−0.170*** (0.030)	−0.171*** (0.030)	−0.171*** (0.026)
PHCD × SUP				−0.154*** (0.028)	−0.153*** (0.025)
Controls	No	Yes	Yes	Yes	Yes
Observations	914	914	914	914	1,247
R²	0.231	0.245	0.273	0.296	0.271

Note: The dependent variable is AI-related career anxiety, standardized. Column (1) omits controls; columns (2)–(4) add controls, support, and the interaction on the matched sample; column (5) repeats column (4) on the full first-wave sample. Values in parentheses are heteroskedasticity-consistent (HC3) standard errors. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

4.4. Second Stage: The Inverted U of Adaptation (H2, H5a)

Table 7 estimates Equation (2). Column (1), the naive linear specification, is exactly the regression a non-systems reading would run, and it is nearly silent: anxiety appears to matter little for adaptation. Column (2) adds the square and the picture inverts: the linear term is 0.174 and the quadratic is −0.237 (SE 0.025), significant at any conventional level, and the within-sample fit improves by 0.099 in R². Columns (3) and (4) add the literacy interaction and the full antecedent set; the curvature is untouched. Column (5) re-estimates

column (4) with standard errors clustered on the twenty occupation groups, anticipating the concern that occupation-level shocks correlate residuals; the conclusion stands, and the wild-cluster bootstrap of Section 4.9 sharpens it.

Table 7. Second stage: the curvilinear conversion of anxiety into adaptation (H2, H5a).

	(1)	(2)	(3)	(4)	(5)
ANX	0.149*** (0.037)	0.152*** (0.031)	0.153*** (0.030)	0.174*** (0.034)	0.174*** (0.030)
ANX ²		−0.266*** (0.027)	−0.256*** (0.026)	−0.237*** (0.025)	−0.237*** (0.020)
LIT			0.278*** (0.030)	0.279*** (0.029)	0.279*** (0.028)
ANX × LIT			0.138*** (0.029)	0.139*** (0.029)	0.139*** (0.028)
SUP				0.165*** (0.031)	0.165*** (0.037)
PHCD				0.009 (0.034)	0.009 (0.037)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	914	914	914	914	914
R ²	0.031	0.130	0.223	0.248	0.248

Note: The dependent variable is occupational adaptation at the second wave, standardized. Columns (1)–(4) use HC3 standard errors; column (5) repeats column (4) with standard errors clustered on the 20 occupation groups. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

Because a negative quadratic coefficient does not by itself establish an interior optimum [28,88], Table 8 reports the composite test. The slope at the first percentile of anxiety is 1.162 ($t = 10.31$); at the 99th percentile it is -0.846 ($t = -7.46$); the Sasabuchi p -value for the composite null of monotonicity is below 0.0001. The turning point sits at 0.37 standard deviations above mean anxiety, with a 95 percent Fieller interval of [0.22, 0.55] that excludes both extremes of the observed range, and 38.5 percent of respondents sit to its right — two in five graduates are already in the region where additional anxiety predicts less adaptation, not more. Figure 1a draws the estimated response with its 95 percent confidence band, the turning point with its interval, and the observed anxiety distribution beneath the axes; the estimate is drawn only over the observed support. H2 is supported. The peak sits 0.37 standard deviations of anxiety above the mean, where predicted adaptation is 0.03 standard deviations of the outcome higher than at mean anxiety; at two standard deviations of anxiety the same prediction is 0.60 lower — a swing of 0.63 standard deviations of the outcome between the optimum and the far tail.

Table 8. Formal validation of the inverted U.

Quantity	Estimate
Slope at the 1st percentile of ANX (−2.09)	1.162 (t = 10.31)
Slope at the 99th percentile of ANX (2.15)	−0.846 (t = −7.46)
Sasabuchi test of the composite null (p)	< 0.0001
Turning point (SD of ANX)	0.368 (SE 0.081)
95% Fieller interval	[0.221, 0.546]
Respondents beyond the turning point (%)	38.5
Turning point at LIT = −1 SD	0.075 (SE 0.095)
Turning point at LIT = +1 SD	0.661 (SE 0.119)
Displacement of the turning point per SD of literacy	0.293 (SE 0.067)

Note: Slopes and tests follow Lind and Mehlum’s composite procedure on the full specification of Table 7, column (4); the turning point interval is a Fieller construction, and the literacy-conditional turning points and their displacement are Equation (4) evaluated on the same specification, with delta-method standard errors. Source: authors’ own calculations.

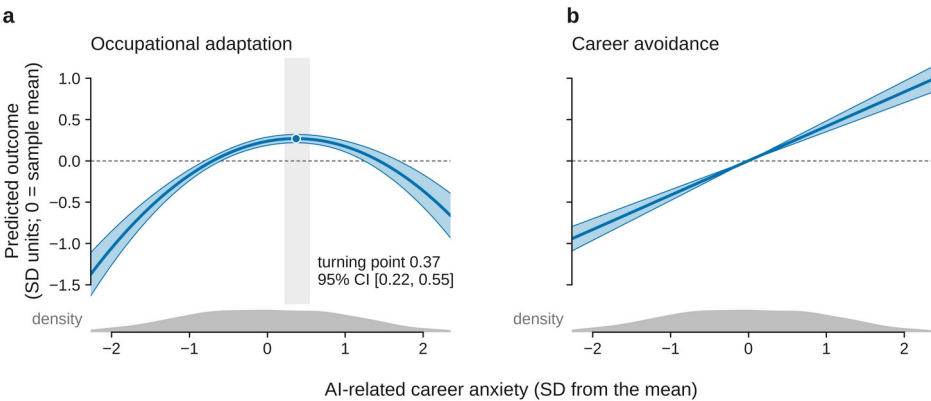


Figure 1. The dual response to AI-related career anxiety: (a) occupational adaptation; (b) career avoidance.

4.5. The Avoidance Channel (H3, H5b)

Table 9 estimates Equation (3), and Figure 1b plots it beside the adaptation response on the same vertical scale. Avoidance rises with anxiety at 0.417 per standard deviation in the full specification, and — the diagnostic contrast with H2 — shows no curvature: a quadratic term added in column (2) is small (0.043) and does not reach the five percent level (p = 0.054). The asymmetry between the two panels of Figure 1 is the paper’s central descriptive fact: the same error signal that buys correction only within a bounded band buys escape without bound. Literacy operates on this channel too: the anxiety-avoidance slope falls by 0.146 per standard deviation of literacy (p < 0.001), supporting H5b.

Table 9. The avoidance channel (H3, H5b).

	(1)	(2)	(3)	(4)	(5)
ANX	0.472*** (0.029)	0.471*** (0.029)	0.469*** (0.028)	0.417*** (0.033)	0.417*** (0.030)
ANX ²		0.043* (0.023)			
LIT			−0.136*** (0.028)	−0.132*** (0.028)	−0.132*** (0.023)
ANX × LIT			−0.148*** (0.029)	−0.146*** (0.029)	−0.146*** (0.029)
SUP				−0.116*** (0.028)	−0.116*** (0.023)
PHCD				0.071** (0.036)	0.071* (0.036)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	914	914	914	914	914
R ²	0.230	0.233	0.269	0.284	0.284

Note: The dependent variable is career avoidance at the second wave, standardized. Column (2) adds the quadratic term, which is insignificant; columns (3)–(4) add literacy and its interaction. Values in parentheses are HC3 standard errors in columns (1)–(4) and occupation-clustered standard errors in column (5). ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

4.6. The Conduit: Conditional Indirect Effects (H4a, H4b)

Table 10 reports the bootstrapped conditional-process estimates of Equation (5). Read the adaptation panel at high support and high literacy: a standard deviation of perceived depreciation transmitted through anxiety raises adaptation by 0.262 when anxiety is low (95 percent CI [0.188, 0.340]), but the same transmission at one standard deviation of anxiety above the mean, at low support and low literacy, is −0.281 ([−0.371, −0.193]) — the identical percept, carried by the identical channel, now demobilizes. The index of curvilinear mediation, $2a_1b_2 = -0.231$ ([−0.286, −0.174]), certifies that the sign reversal is a property of the transmitted effect and not a boundary artifact; the indices of moderated mediation are −0.027 ([−0.042, −0.014]) for support in the first stage and 0.068 ([0.040, 0.096]) for literacy in the second [27]. The avoidance channel is uniformly positive — 0.274 at low literacy, 0.132 at high — and its literacy index is −0.071 ([−0.099, −0.043]). H4a and H4b are supported, with the qualification H4a itself predicted: the conduit’s sign depends on where in the anxiety distribution the graduate stands.

Table 10. Conditional indirect effects of perceived depreciation, through anxiety (H4a, H4b).

Path and conditioning values	Estimate	95% bootstrap CI
SUP = −1, ANX = −1, LIT = −1	0.326	[0.231, 0.422]
SUP = −1, ANX = −1, LIT = +1	0.504	[0.404, 0.609]
SUP = −1, ANX = +0, LIT = −1	0.023	[−0.034, 0.078]
SUP = −1, ANX = +0, LIT = +1	0.201	[0.143, 0.260]
SUP = −1, ANX = +1, LIT = −1	−0.281	[−0.371, −0.193]
SUP = −1, ANX = +1, LIT = +1	−0.103	[−0.189, −0.015]
SUP = +1, ANX = −1, LIT = −1	0.169	[0.110, 0.235]
SUP = +1, ANX = −1, LIT = +1	0.262	[0.188, 0.340]
SUP = +1, ANX = +0, LIT = −1	0.012	[−0.018, 0.041]
SUP = +1, ANX = +0, LIT = +1	0.104	[0.069, 0.143]
SUP = +1, ANX = +1, LIT = −1	−0.146	[−0.205, −0.094]
SUP = +1, ANX = +1, LIT = +1	−0.053	[−0.103, −0.008]
Avoidance path, LIT = −1	0.274	[0.226, 0.325]
Avoidance path, LIT = +1	0.132	[0.090, 0.176]
IMM of SUP (first stage, adaptation)	−0.027	[−0.042, −0.014]
IMM of LIT (second stage, adaptation)	0.068	[0.040, 0.096]
Index of curvilinear mediation (2·a1·b2)	−0.231	[−0.286, −0.174]
IMM of LIT (second stage, avoidance)	−0.071	[−0.099, −0.043]

Note: Instantaneous indirect effects $\omega(m, w, l)$ of Equation (5), with SUP, ANX and LIT set to the stated values in standard deviations. Confidence intervals are percentile intervals from 5,000 bootstrap resamples of individuals; the seed is fixed in the released code. IMM denotes an index of moderated mediation. Source: authors’ own calculations.

4.7. Boundary Conditions in the Loop (H5a, H6)

Figure 2 assembles the two boundary conditions. Panel a draws the adaptation response at literacy one standard deviation below and above the mean: the low-literacy curve peaks at 0.07 standard deviations of anxiety, the high-literacy curve at 0.66, a displacement of 0.29 standard deviations per standard deviation of literacy (SE 0.07, $p < 0.001$) [28]. Literacy does not flatten the curve — a quadratic-by-literacy term is small and insignificant ($p = 0.434$) — it moves the peak, which is precisely the actuator-range prediction: capability extends the band over which the error signal remains productive rather than changing the physics of saturation. Panel b draws the first stage across support with its confidence band and the observed support distribution, the visual counterpart of Table 6, column (4). Both panels annotate their key quantities directly on the drawing, and neither uses a legend, so nothing occludes the estimates.

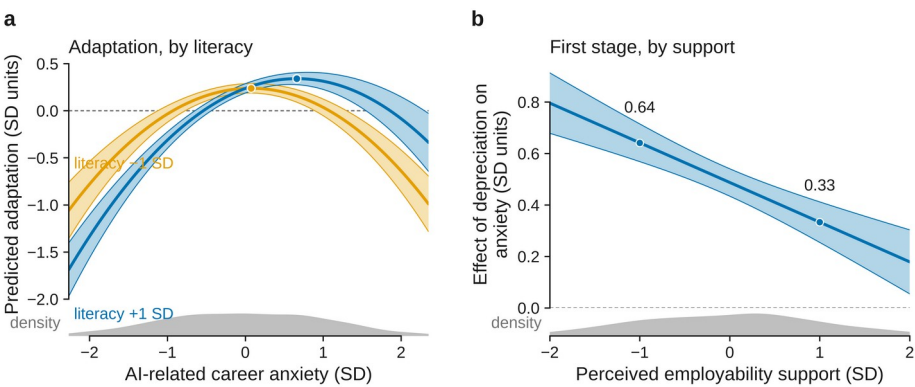


Figure 2. Boundary conditions of the control loop: (a) the adaptation response at low and high generative-AI literacy; (b) the first stage across employability support.

4.8. Heterogeneity: A Property of the Cohort, Not of a Stratum

Table 11 re-estimates the full adaptation model within ten subsamples defined by gender, major, city tier, occupational exposure and employment status, and Figure 3a plots the marginal effect of anxiety at the sample mean for each with 95 percent confidence intervals. The curvature is negative and significant in every subsample, and none of the five between-group differences in the marginal effect approaches significance (the smallest p-value is 0.492). The inverted U is not carried by women rather than men, by STEM graduates rather than humanists, or by the exposed rather than the sheltered; it is a property of the cohort. For a mechanism claimed to be structural — saturation of a control signal — this uniformity is itself evidence, and it forecloses the objection that the curve is an artifact of pooling heterogeneous linear responses.

Table 11. Subsample estimates: the mechanism across the cohort.

Subsample	N	dADP/dANX at mean	(SE)	ANX ²
Female	483	0.187	(0.048)	−0.260***
Male	431	0.166	(0.052)	−0.205***
difference (z, p)		0.29	0.774	
STEM	368	0.200	(0.055)	−0.228***
Non-STEM	546	0.161	(0.046)	−0.244***
difference (z, p)		0.55	0.586	
Tier-1 city	298	0.209	(0.062)	−0.240***
Other cities	616	0.158	(0.042)	−0.231***
difference (z, p)		0.69	0.492	
High exposure	477	0.180	(0.046)	−0.213***
Low exposure	437	0.164	(0.056)	−0.255***
difference (z, p)		0.21	0.831	
Employed	676	0.181	(0.038)	−0.238***
Not employed	238	0.171	(0.083)	−0.224***
difference (z, p)		0.11	0.914	

Note: Each pair of rows re-estimates the full adaptation model within the stated subsample; the third column is the marginal effect of anxiety at the sample mean, the final column the subsample quadratic coefficient. Difference rows report a two-sample z-test of the marginal effects. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

4.9. Robustness

Table 12 subjects the curvilinear second stage to six replications. (1) Replacing the adaptation index with the log of reported reskilling hours — a behavioral count, not a Likert judgment — returns a quadratic of -0.140 ($p < 0.001$). (2) Weighting the retained sample by inverse retention probabilities estimated from all first-wave observables leaves the quadratic at -0.237 . (3) Dropping the software and information-technology occupations, whose graduates might mechanically both worry and reskill, gives -0.235 . (4) Unit-weighted composites in place of factor scores give -0.237 [86]. (5) Partialling the marker score out of every construct gives -0.240 [80]. (6) The wild-cluster bootstrap over the twenty occupation groups, the appropriate small-G inference [91], puts the curvature’s p-value at 0.001. The inverted U survives every change of measure, sample, weighting and inference.

Table 12. Robustness of the curvilinear second stage.

Specification	ANX	(SE)	ANX ²	(SE)	N
(1) log reskilling hours	0.090***	(0.029)	−0.140***	(0.022)	914
(2) inverse-probability weighted	0.174***	(0.029)	−0.237***	(0.021)	914
(3) excluding software and IT	0.170***	(0.035)	−0.235***	(0.027)	838
(4) unit-weighted composites	0.174***	(0.034)	−0.237***	(0.025)	914
(5) marker-partialled scores	0.178***	(0.034)	−0.240***	(0.025)	914
(6) wild cluster bootstrap p for ANX ² (20 occupation clusters)			0.001		914

Note: Each row replicates the full adaptation specification under the stated change. Row (2) weights by inverse retention probabilities from a first-wave linear probability model; row (6) reports the wild-cluster bootstrap p-value for the quadratic term with Rademacher weights and the null imposed, 999 replications. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

4.10. The Configurational Test (H7a, H7b)

Table 13 reports the calibration anchors and the necessity analysis. No condition, present or absent, reaches the 0.90 consistency conventionally required of a necessary condition; the maximum anywhere is 0.708. H7a is supported: there is no single gate through which every adapting graduate passes.

Table 13. Fuzzy-set calibration and necessity analysis.

Condition	Full (0.95)	Crossover (0.50)	Out (0.05)
Panel A: calibration anchors (raw score at membership 0.95 / 0.50 / 0.05)			
PHCD	1.70	0.03	−1.75
ANX	1.58	0.03	−1.68
LIT	1.63	0.03	−1.76
SUP	1.64	0.01	−1.61
EXPO	1.56	0.09	−1.89
Panel B: necessity analysis for high adaptation			
Condition	Consistency	Coverage	
PHCD	0.665	0.667	
~PHCD	0.639	0.647	
ANX	0.674	0.677	
~ANX	0.614	0.621	
LIT	0.708	0.718	
~LIT	0.592	0.593	
SUP	0.682	0.681	
~SUP	0.605	0.615	
EXPO	0.645	0.659	
~EXPO	0.647	0.643	

Note: Direct-method calibration with anchors at the 95th, 50th and 5th sample percentiles of each standardized score. ~ denotes negation. No consistency reaches the 0.90 threshold conventionally required of a necessary condition. Source: authors’ own calculations.

Sufficiency is a different story, and it is the sharpest result in the paper. Of the 31 truth-table rows that clear the frequency threshold, only 4 pass both consistency screens, and they minimize to a single intermediate solution — identical to the parsimonious solution, so every condition in it is core [30]: the conjunction of anxiety, literacy and support (Table 14), with consistency 0.892 and coverage 0.406. Figure 3b plots each respondent’s membership in this recipe against membership in high adaptation; the lower-right triangle, where recipe members fail to adapt, is nearly empty, which is what sufficiency looks like case by case. The near-misses are as informative as the solution: anxiety-with-literacy-but-without-support configurations reach raw consistencies near 0.91 yet fail the proportional-reduction screen, and the highest-consistency row for the negation of adaptation (raw consistency 0.893) fails the proportional-reduction screen (PRI 0.602 < 0.70), so no configuration is reliably sufficient for failing to adapt. H7b is therefore not supported, and its failure is the finding: the system offers one road to adaptation, not many. Anxiety appears in the recipe — the signal is an ingredient of correction — but it never suffices without the actuator and the environment, which is the conjunction the control-loop reading predicted [24,25].

Table 14. Sufficient configuration for high occupational adaptation.

	PHCD	ANX	LIT	SUP	EXPO	Consistency	Raw cov.	Unique cov.
C1		●	●	●		0.892	0.406	0.406
Solution						0.892	0.406	

Note: ● indicates the presence of a condition and ⊗ its absence; a blank cell means the condition is irrelevant to the configuration. All marked conditions are core: the parsimonious and intermediate solutions coincide. Truth-table thresholds: frequency ≥ 10 cases, raw consistency ≥

0.80, PRI $\geq$ 0.70. No configuration passes the same thresholds for the negation of adaptation.
Source: authors' own calculations.

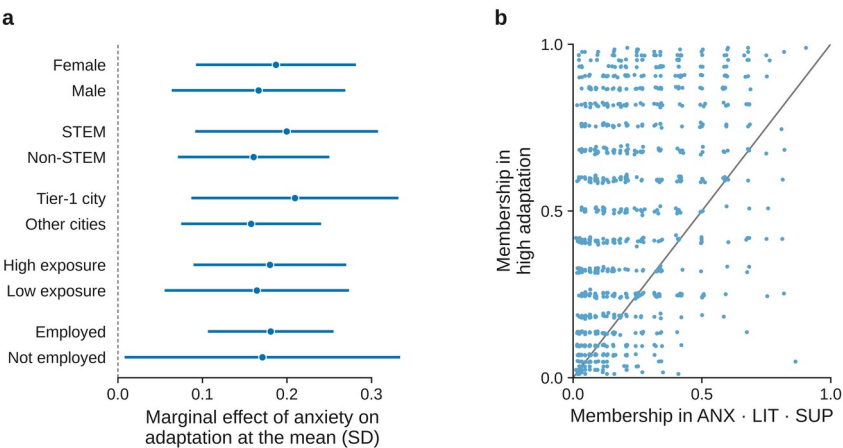


Figure 3. Generality and sufficiency: (a) the marginal effect of anxiety on adaptation by subsample; (b) case-level sufficiency of the single configuration.

5. Discussion

5.1. Theoretical Implications

The first implication concerns how the literature on artificial intelligence and work should model the displaced margin. Adjustment models typically assume that displacement pressure translates smoothly into retraining uptake [13,14]. The present estimates say the translation is nonmonotonic at the individual level: pressure mobilizes up to a point that forty percent of the entry cohort has already passed, and demobilizes beyond it. Aggregate retraining elasticities estimated on calm populations will overpredict the adjustment of anxious ones, and the error grows with exactly the exposure that makes adjustment urgent [1,4]. Early-career scarring research [15,18] gains a psychological microfoundation: part of what converts a bad entry draw into a persistent disadvantage may be the saturation of the very signal that should drive recovery.

The second implication is for the artificial-intelligence anxiety literature, which has largely treated anxiety as an outcome to be predicted or a hazard to be reduced [19,20,34]. Treating it instead as the error signal of a cybernetic loop [24,42] yields predictions that a valence-only reading does not: the interior optimum (H2), the actuator-dependence of the optimum's location (H5a), and the conjunctural sufficiency (H7). All three survived their tests. The point is not that anxiety is good or bad but that it is informative, and that the information is wasted — or turned destructive — by a system that supplies no means of acting on it. This reframing connects the psychology of technological threat to Ashby's requisite variety [25] and to the capability literatures in organizational learning [92–94], which have long held that signals without absorption capacity produce noise, not learning.

The third implication is methodological. The configurational result — one recipe, all conditions core, no recipe for the negation — is unusual in applied fuzzy-set work, where multiple configurations are the norm and are routinely read as equifinality [30,31]. The paper shows that a tight theoretical loop can predict, and data can deliver, the opposite: conjunctural necessity of complements around a single path. Reporting the failure of equifinality as a

finding, rather than forcing a multi-recipe reading, is the more faithful use of the method [32,78].

5.2. Practical Implications

For universities, the estimates relocate the lever. Career services that concentrate on reassurance address the sensor; the binding constraints are the actuator and the environment. The literacy moderation implies that embedding generative-AI practice into every curriculum — not as ethics seminars but as supervised, verifiable use in domain work [7,62] — buys a double dividend: it raises adaptation directly and it extends the anxiety range over which graduates keep adapting rather than freezing. The support buffering implies that visible, credible transition infrastructure — placement guarantees, employer-linked reskilling pathways, alumni networks that answer — lowers the anxiety produced by any given level of objective threat, which matters most for the two-fifths of graduates already past the turning point.

For employers, the same logic applies to the junior pipeline they are currently thinning [4,8]. A firm that hires fewer entrants and tells the remainder to “upskill or else” is raising the error signal while shrinking the actuator set — the exact combination the estimates identify as producing avoidance, quiet disengagement and exit contemplation. The configurational result gives the design rule in one line: anxiety converts into adaptation only in the presence of literacy and support, so any communication of threat should arrive bundled with tool access, training time and a visible internal pathway [95,96]. For policymakers confronting record graduate cohorts [10,45,97,98], the estimates argue for shifting marginal spending from awareness campaigns, which raise the signal, toward subsidized generative-AI credentialing and employer-linked placement, which complete the loop; and for targeting that spending on the high-anxiety segment, where the same yuan buys the largest swing from demobilization back to adaptation.

5.3. A Systems Reading of the Whole

Assembled, the estimates describe a socio-technical control system operating near its limit. The disturbance — task-level displacement of entry work — enters through a perception whose gain is set by support (Table 6). The error signal saturates: beyond one third of a standard deviation of anxiety, additional signal produces less correction (Table 8), and the share of the cohort past that point is large. Capability moves the saturation boundary outward but does not remove it (Figure 2a). The corrective channel and the escape channel run in parallel, and the escape channel never saturates (Figure 1b). Finally, the system is conjunctural: correction is produced by one specific combination of signal, capability and environment, and by nothing else (Table 14). None of these properties is visible to a linear, single-equation, single-condition analysis; each is the kind of emergent, feedback-borne behavior the systems tradition trained us to look for [40,99–101], and the cohort’s own reading of the moment is a live exercise in collective sensemaking under threat [102]. The entry labor market of the generative-AI era is not a queue that clears; it is an adaptive system whose youngest agents are its most sensitive sensors, and whether their sensitivity ends as adjustment or as scarring depends on infrastructure the sensors do not control.

6. Limitations and Future Research

Five limitations bound the claims. First, although the outcomes are separated from their antecedents by six months and the attrition diagnostics are clean, the design remains observational; unmeasured traits such as dispositional neuroticism could raise both anxiety and behavioral reports, and only a design that manipulates the actuators — for instance, randomized literacy training layered on measured anxiety — can close that door. Second, all psychological constructs are self-reported; the behavioral hours measure and the marker and method-factor diagnostics bound the resulting bias but do not eliminate it [85]. Third, the sample is Chinese graduates within three years of graduation, observed at one moment of one technological transition; the loop's parameters — the location of the turning point above all — are surely context-dependent, and comparative replication across institutional regimes with different transition infrastructures [45,103] is the obvious next step. Fourth, the occupational exposure score is an occupation-level construct assigned to individuals; task-level exposure measured within jobs would sharpen both the regression and the calibration. Fifth, the two-wave design identifies one traversal of the loop, not its dynamics: whether adaptation feeds back to lower depreciation perceptions, closing a balancing loop, or whether avoidance compounds into the reinforcing spiral that conservation-of-resources theory predicts [41,65], requires a multi-wave panel analyzed with the estimation machinery the system-dynamics tradition has built for exactly such loops [104].

7. Conclusions

This paper asked how the first graduate cohort of the generative-AI era converts the perception of depreciating human capital into early-career choices. The answer is a control loop with a saturating signal: depreciation becomes anxiety at a rate set by support; anxiety becomes adaptation only within a bounded band that two-fifths of the cohort has already left, while it becomes avoidance without bound; generative-AI literacy extends the productive band; and only the full conjunction of signal, capability and support is sufficient for adaptation, with no alternative recipe and no recipe for its negation. The finding reframes what is usually called an anxiety problem as an infrastructure problem. A generation's error signal is already loud; whether it ends as occupational adaptation or as scarring depends on whether universities, employers and governments supply the actuators that let a loud signal steer.

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Conflicts of Interest: The authors declare no conflicts of interest.

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